	UNIT 1
1	Which device converts solar energy directly into electrical energy? A) Solar collector B) Solar cooker C) Solar cell D) Heat exchanger
2	A fuel cell produces electricity by A) Nuclear reaction B) Electrochemical reaction C) Combustion D) Friction
3	Non-conventional energy sources help in A) Increasing pollution B) Depleting resources C) Sustainable development D) Increasing fossil fuel use
4	The efficiency of a solar cell is generally in the range of: A) 5–10% B) 15–25% C) 50–60% D) 80–90
5	A fuel cell produces electricity by A) Nuclear reaction B) Electrochemical reaction C) Combustion D) Friction
6	The main reason for using non-conventional energy sources is A) To increase pollution B) To conserve fossil fuels C) To reduce electricity use D) To increase coal demand
7	Which of the following is used to store solar energy? A) Battery B) Turbine C) Boiler D) Condense
8	Hydrogen can be produced by A) Electrolysis of water B) Burning coal C) Wind turbine D) Distillation of oil
9	Which energy source is most suitable for remote villages? A) Diesel generator B) Solar PV system C) Coal plant D) Thermal plant
10	The unit of solar radiation is A) kWh/m^2 B) m/s C) Pa D) kg/m^3
11	Fuel cell converts ________________________________ A) Chemical energy into electrical energy B) Solar energy into sound energy C) Mechanical energy into heat energy D) Wind energy into heat energy

12	Which of the following has the least environmental impact? A) Coal B) Diesel C) Solar D) Petrol
13	Main product of hydrogen combustion is A) CO_2 B) SO_2 C) H_2O D) CO
14	A solar pond is used to A) Generate wind energy B) Store solar thermal energy C) Produce coal D) Measure radiation
15	Which of the following is a non-conventional energy source? A) Coal B) Petroleum C) Solar energy D) Natural gas
16	Which is known as fuel of the future? A) Coal B) Petrol C) Hydrogen D) Diesel
17	The tilt angle of a solar collector depends on A) Rainfall B) Latitude of location C) Wind speed D) Humidity
18	Solar energy is converted into thermal energy by A) Solar cell B) Solar collector C) Battery D) Inverter
19	The most commonly used material in solar cells is A) Copper B) Silicon C) Iron D) Aluminium
20	The efficiency of a solar cell is generally in the range of A) 5–10% B) 15–25% C) 50–60% D) 80–90%
21	The main advantage of non-conventional energy sources is A) High pollution B) Limited availability C) Eco-friendly nature D) High operating cost
22	Which of the following is a disadvantage of a solar PV system? A) No maintenance B) Silent operation C) High initial cost D) No pollution
23	The major disadvantage of solar energy is A) No maintenance B) Pollution C) Intermittent availability D) High noise
24	Solar energy is converted into thermal energy by

	A) Solar cell B) Solar collector C) Battery D) Inverter
25	Hydrogen has A) High calorific value B) Low calorific value C) No energy D) Fixed energy only
26	Which of the following is a major application of fuel cell? A) Spacecraft B) Coal washing C) Textile industry D) Cement making
27	Hydrogen is considered as A) Primary fuel B) Secondary fuel C) Fossil fuel D) Solid fuel
28	Which device converts solar energy directly into electrical energy? A) Solar collector B) Solar cooker C) Solar cell D) Heat exchanger
29	Fuel cell converts ___________________________________ A) Chemical energy into electrical energy B) Solar energy into sound energy C) Mechanical energy into heat energy D) Wind energy into heat energy
30	A solar inverter converts A) AC to DC B) DC to AC C) Heat to electricity D) Wind to electricity

	UNIT 2
31	Wind energy is converted into electrical energy by A) Compressor B) Turbine C) Transformer D) Boiler
32	Which of the following affects wind power generation? A) Wind speed B) Air density C) Rotor swept area D) All of the above
33	A detailed energy audit is also known as A)Walk-through audit B)Preliminary audit C)Investment-grade audit D) Spot audit

34	Energy audits are generally classified into A) Electrical and mechanical only B)Preliminary and detailed C) Indoor and outdoor D) Renewable and non-renewable
35	The device used to measure wind speed is A) Barometer B) Anemometer C) Thermometer D) Hygrometer
36	The device used to measure wind speed is A) Barometer B) Anemometer C) Thermometer D) Hygrometer
37	Which is an example of biofuel? A)Petrol B) Diesel C) Ethanol D) Coal
38	The main drawback of wind energy is A) High pollution B) Noise and variable output C) Fuel shortage D) Radioactivity
39	Biogas plant works on A) Combustion B) Anaerobic digestion C) Distillation D) Cracking
40	Vertical axis wind turbine is suitable for A) High altitude only B) Low wind speed C) Nuclear plants D) Solar farms
41	OTEC uses A) Wind difference B) Pressure difference C) Temperature difference in ocean water D) Salinity difference only
42	Methane is mainly emitted from A)Solar panels B) Livestock and landfills C)Wind turbines D) Hydropower stations
43	OTEC uses A) Wind difference B) Pressure difference C) Temperature difference in ocean water D) Salinity difference only
44	Energy efficiency is defined as A) Energy output greater than input B)Using less energy for the same output C) Maximum fuel burning D) Energy wastage
45	Which protocol is widely used for carbon accounting? A) HTTP B) GHG Protocol C) SMTP D) FTP
46	Which energy-saving measure is most suitable for lighting systems?

	A) Increase voltage B)Replace LEDs with incandescent bulbs C) Install LED lamps D) Use larger transformers
47	Carbon footprint is usually measured in A) Litters B)Joules C) Tons of CO_2 equivalent D)Hertz
48	What is the primary objective of an energy audit? A) Increase energy consumption B) Identify energy-saving opportunities C) Replace all electrical equipment D) Reduce employee workload
49	Which instrument is commonly used to detect heat loss in buildings? A)Lux meter B)Tachometer C) Thermal imaging camera D)pH meter
50	Carbon footprint is usually measured in A) Litres B) Joules C) Tons of CO_2 equivalent D) Hertz
51	The power available in wind is proportional to A) Wind speed B) Square of wind speed C) Cube of wind speed D) Fourth power of wind speed
52	Energy efficiency is defined as A. Energy output greater than input B. Using less energy for the same output C. Maximum fuel burning D. Energy wastage
53	Energy efficiency is defined as A) Energy output greater than input B) Using less energy for the same output C) Maximum fuel burning D) Energy wastage
54	The Betz limit of wind turbine is approximately A) 25% B) 40% C) 59.3% D) 80%
55	The minimum wind speed required for power generation is approximately: A) 2 m/s B) 5 m/s C) 20 m/s D) 50 m/s
56	The Betz limit of wind turbine is approximately

	A) 25% B) 40% C) 59.3% D) 80%
57	The Betz limit of wind turbine is approximately A) 25% B) 40% C) 59.3% D) 80%
58	The process of converting biomass into charcoal is called A) Digestion B) Carbonization C) Fermentation D) Gasification
59	Which gas contributes most to global warming? A) Oxygen B) Nitrogen C) Carbon dioxide D) Helium
60	Power factor improvement helps to A) Increase fuel use B) Reduce electricity losses C) Increase harmonic distortion D) Stop transformer operation